

e-CAM82_CUOAGX

Getting Started Manual



Version 1.4

e-con Systems

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Disclaimer

The specifications of e-CAM82_CUOAGX board and instructions on how to use this board with NVIDIA® Jetson AGX ORIN™/ Xavier™ development kits are provided as reference only and e-con Systems reserves the right to edit/modify this document without any prior intimation of whatsoever.

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Introduction to e-CAM82_CUOAGX

e-CAM82_CUOAGX is 8 MP customer lens camera module designed and developed by e-con Systems, a leading Embedded Product Design Services Company which specializes in advanced camera solutions. It can be interfaced to the J509 camera connector of the NVIDIA® Jetson AGX ORIN™ / Xavier™ development kit using the e-CAM80_CUMI334_ADAP board.

e-CAM82_CUOAGX is provided with CS-mount lens holder. e-CAM82_CUOAGX is a color camera that delivers RGB RAW data. The supported resolutions and frame rates are listed in following table.

Table 1: Maximum Frame Rate Supported for e-CAM82_CUOAGX

Platform	Lanes	Resolution	Frame Rate(fps) in 10-bit Output	Frame Rate(fps) in 12-bit Output
NVIDIA® Jetson AGX ORIN™ / Xavier™	4-lane	1920 x 1080	90	90
		3840 x 2160	60	50
	2-lane	1920 x 1080	62	62
		3840 x 2160	39	33

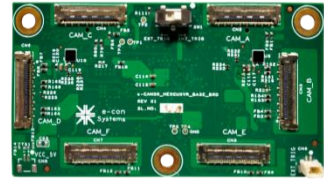
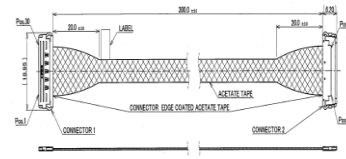
This document describes how to interface the e-CAM82_CUOAGX board on NVIDIA® Jetson AGX ORIN™ / Xavier™ development kit and how to use the e-CAM82_CUOAGX board.

Parts Supplied

The following table lists the parts supplied with the kit.

Table 2: Parts Supplied and its Quantities.

Parts Supplied	Images	Quantity
Custom lens camera module (e-CAM84_CUMI485C_MOD_H02R1)		1

Adaptor board (e-CAM80_CUMI334_ADP)		1
Adaptor board (e-CAM30_HEXCUXVR_BASE_BRD)		1
30cm IPEX cable		1
Lens		1

Description

NVIDIA® Jetson AGX ORIN™ / Xavier™ are newly launched small size, low power, artificial intelligence (AI) system-based evaluation boards, developed by NVIDIA®. NVIDIA® Jetson AGX ORIN™ / Xavier™ development kits are having two individual 4-lane MIPI CSI-2 camera connection options. e-CAM82_CUOAGX uses these 4-lane MIPI CSI-2 interface for connecting 8 MP camera modules.

e-CAM82_CUOAGX is a multi-board solution, which has two boards as follows:

- Camera module (e-CAM84_CUMI485C_MOD_H02R1)
- Adaptor board (e-CAM80_CUMI334_ADP)

The camera module is a small, low-power, high performance 8 MP camera with the external ISP. It is based on 1/1.2" IMX485 sensor from STARVIS family of CMOS image sensor from SONY. The IMX485 is a 1/1.2" optical form-factor, CMOS image sensor with a variable speed shutter.

The front and rear views of e-CAM84_CUMI485C_MOD_H02R1 board and e-CAM80_CUMI334_ADP adaptor board are shown in following figures.

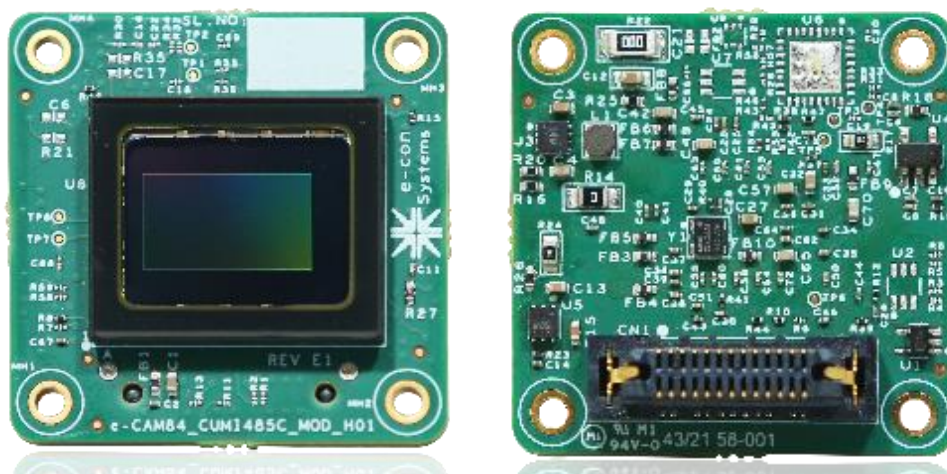


Figure 1: Front and Rear Views of e-CAM84_CUMI485C_MOD_H02R1 Board



Figure 2: Front and Rear Views of e-CAM80_CUMI334_ADP Adaptor Board

The e-CAM82_CUOAGX camera module has one 26-pin Samtec connector (CN1) for mating with e-CAM80_CUMI334_ADP adaptor board. This adaptor board acts as a bridge between the camera module and the NVIDIA® Jetson AGX ORIN™ / Xavier™ development kit. It supplies the voltage required for camera module. e-CAM82_CUOAGX adaptor board consists of 30-pin IPEX connector (CN2), through which e-CAM82_CUOAGX is connected to NVIDIA® Jetson AGX ORIN™ / Xavier™ development kit using the 30 cm IPEX cable.

e-CAM82_CUOAGX Board Handling Procedure

This section describes the handling procedure of e-CAM82_CUOAGX board.

The procedure to assemble camera board to NVIDIA® Jetson AGX ORIN™ / Xavier™ carrier board is described in the following sections:

- [Camera board connection](#)
- [Interfacing with the NVIDIA® Jetson AGX ORIN™ development kit](#)
- [Interfacing with NVIDIA® Jetson AGX Xavier™ development kit](#)

Camera Board Connection

e-CAM82_CUOAGX is provided with 30cm IPEX cable, for connecting with NVIDIA® Jetson AGX ORIN™ / Xavier™ camera boards.

The conductive region and locking region are shown in following figure.

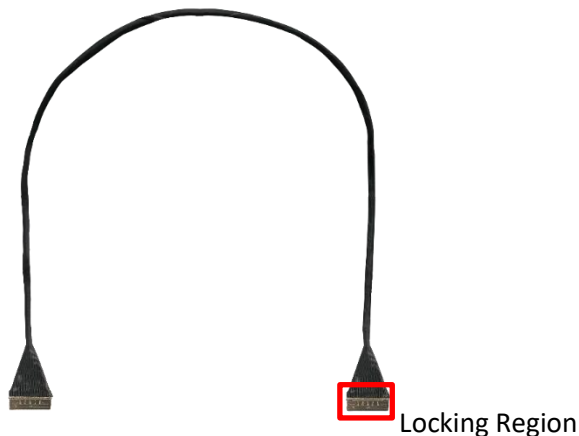


Figure 3: IPEX Cable Conductive and Locking Side Location

The steps to connect the camera board are as follows:

1. Connect IPEX cable at CN2 connector of adapter board and lock it.

IPEX cable has a lock bar, which is used for locking and unlocking the cable as shown in following figure.

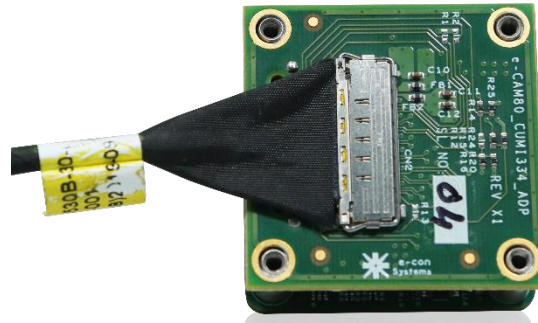


Figure 4: CN2 Connector Location

2. Connect the module board to adapter board through board-to-board Samtech connector CN1 as shown in following figure.



Figure 5: e-CAM82_CUOAGX CN2 Connector Locked with Cable

3. Keep the cable connected board combination in enclosure.

Note: Care must be taken, while connecting cable to e-CAM82_CUOAGX connector and keeping combination in enclosure.



Figure 6: e-CAM82_CUOAGX with Enclosure

4. Mount custom lens to enclosure as shown in the following figure.



Figure 7: Custom Lens mounted to Enclosure

Interfacing with NVIDIA Jetson AGX ORIN Development Kit

The steps to interface with the NVIDIA® Jetson AGX ORIN™ development kit are as follows:

1. Connect base board on to camera connector of NVIDIA® Jetson AGX ORIN™ development kit.

Base board has board to board connector as shown in following figure.



Figure 8: Base Board on NVIDIA Jetson AGX ORIN

2. Connect IPEX cable at any of the six connectors in e-CAM30_HEXCUXVR_BASE_BRD base board and lock it as shown in following figure.



Figure 9: IPEX Connector Location

Note: Care must be taken, while connecting cable to camera connector of NVIDIA® Jetson AGX ORIN™ development kit.

Now, e-CAM82_CUOAGX is connected to NVIDIA® Jetson AGX ORIN™ development kit as shown in following figure.



Figure 10: e-CAM82_CUOAGX Setup Interfaced to NVIDIA Jetson AGX ORIN Development Kit

NVIDIA® Jetson AGX ORIN™ development kit can be powered either from USB Micro-B connector or DC power jack connector. Since USB is used for low power applications, e-CAM82_CUOAGX supports only low resolutions in this power supply. Connecting 5V 3.3A power supply to DC jack will be the ideal solution, when using e-CAM82_CUOAGX with NVIDIA® Jetson AGX ORIN™ development kit to use its full listed resolutions.

The power supply connectors in NVIDIA® Jetson AGX ORIN™ development kit is shown in following figure.



Figure 11: Power Supply Connectors in NVIDIA Jetson AGX ORIN Development Kit

e-CAM30_HEXCUXVR_BASE_BRD base board connected to NVIDIA® Jetson AGX ORIN™ carrier board has six camera connectors where four 4-lane cameras can be connected as shown below.

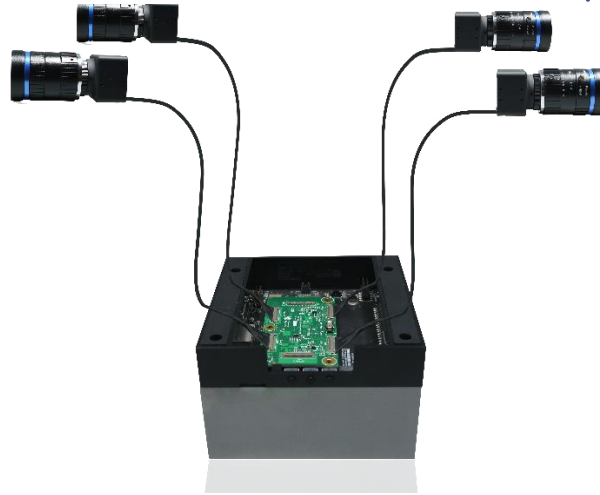


Figure 12: Four Camera Connectors connected to NVIDIA Jetson AGX ORIN Development Kit

Interfacing with NVIDIA Jetson AGX Xavier Development Kit

The steps to interface with the NVIDIA® Jetson AGX Xavier™ development kit are as follows:

3. Connect base board on to camera connector of NVIDIA® Jetson AGX Xavier™ development kit.

Base board has board to board connector as shown in following figure.



Figure 13: Base Board on NVIDIA Jetson AGX Xavier

4. Connect IPEX cable at any of the six connectors in e-CAM30_HEXCUXVR_BASE_BRD base board and lock it as shown in following figure.



Figure 14: IPEX Connector Location

Note: Care must be taken, while connecting cable to camera connector of NVIDIA® Jetson AGX Xavier™ development kit.

Now, e-CAM82_CUOAGX is connected to NVIDIA® Jetson AGX Xavier™ development kit as shown in following figure.

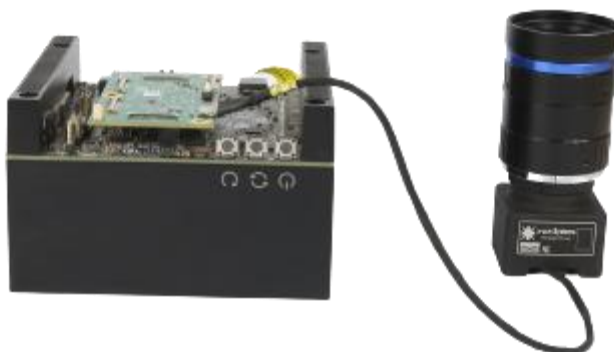


Figure 15: e-CAM82_CUOAGX Setup Interfaced to NVIDIA Jetson AGX Xavier Development Kit

NVIDIA® Jetson AGX Xavier™ development kit can be powered either from USB Micro-B connector or DC power jack connector. Since USB is used for low power applications, e-CAM82_CUOAGX supports only low resolutions in this power supply. Connecting 5V 3.3A power supply to DC jack will be the ideal solution, when using e-CAM82_CUOAGX with NVIDIA® Jetson AGX Xavier™ development kit to use its full listed resolutions.

The power supply connectors in NVIDIA® Jetson AGX Xavier™ development kit is shown in following figure.



Figure 16: Power Supply Connectors in NVIDIA Jetson AGX Xavier Development Kit

Extracting and Flashing e-con provided Binaries

This section describes the extracting and flashing of e-con Systems provided binaries to Jetson AGX Xavier™ / ORIN™ development kit.

The commands and output messages in this manual are represented by different colors as shown in below table.

Table 3: Notation of Colors

Color	Notation
Blue	Commands running in Development PC
Red	Output message in Development PC

Before extracting and flashing the e-con provided binaries in Jetson AGX Xavier™ / ORIN™ development kit, flash **the Jetpack 5.1** provided by Nvidia®, using the SDK manager. Download the SDK manager using <https://developer.nvidia.com/nvidia-sdkmanager> link.

Sign up to an account in NVIDIA developer site to use the SDK manager.

Install the SDK Manager in the host PC using the below command.

```
sudo apt-get install ./sdkmanager  
[version].[build#].deb
```

After installing the SDK manager in the host PC, follow the instructions in the link <https://docs.nvidia.com/sdk-manager/install-with-sdkm-jetson/index.html>.

Copy the release package into the HOME Directory of the flashed Jetson AGX Xavier™/ ORIN™ development kit.

To extract and flash the binaries, follow these steps:

1. Run the following commands to extract the release package in the Jetson™ development kit to obtain the binaries

```
tar -xaf e-  
CAM82_CUOAGX_JETSON_XAVIER_ORIN_<L4T>_<RELEASE_DATE>_<  
VERSION >.tar.gz  
  
cd e-  
CAM82_CUOAGX_JETSON_XAVIER_ORIN_<L4T>_<RELEASE_DATE>_<  
VERSION >
```

The folder contains the necessary tools to immediately flash the binaries in Jetson ORIN™ / Xavier™ with the kernel, camera drivers, and application.

2. Run the following commands in the Jetson™ kit.

```
sudo chmod +x install_binaries.sh
sudo -E ./install_binaries.sh
```

3. After running the above command in the terminal, you can see the command like below,

```
nvidia@nvidia-desktop:~/e-CAM82_CUOAGX_JETSON_XAVIER_ORIN_L4T35.2.1_13-MAR-2023_R02$ sudo -E ./install_binaries.sh
Enter Lane Configuration mode to flash
    1. 2lane
    2. 4lane

```

You can enter LANE configuration mode to flash the dtb file.

eg. Enter “1” or “2lane” in terminal to flash the 2-lane mode configuration.

Note: Entering invalid input will exit from the flashing.

This product package supports both 2 Lane and 4 Lane configurations. During the binary installation, the script will halt at the device tree installation stage and displays the options available.

Please select option 1 to configure the camera in 2lane mode and option 2 to configure the camera in 4lane mode. After the successful installation, the board will reboot and the Jetson™ development kit is ready to use with e-CAM82_CUOAGX.

The steps to be followed in the development board for launching the application are as follows:

1. Run the following command to check whether all the four cameras are initialized.

```
sudo dmesg | grep "Detected eimx485 sensor"
```

The output message for a quad camera setup appears as shown below.

```
Detected eimx485 sensor
Detected eimx485 sensor
Detected eimx485 sensor
Detected eimx485 sensor
```

The output message indicates that all cameras are initialized properly.

2. Run the following command to check the presence of four video nodes.

```
ls /dev/video*
```

The output message for a quad camera setup appears as shown below.

```
video0
video1
```

```
video2
Video3
```

3. Run the following command to set the power mode to maximum for better performance.

```
$ sudo nvpmodel -m 0
```

4. Run the following Jetson clocks and max-clocks shell scripts before launching the eCAM_argus_camera application in the Jetson AGX Xavier™/ ORIN™ development board.

```
$ sudo jetson_clocks
$ sudo /home/max-isp-vi-clks.sh
```

5. Run the following command to launch the sample camera application.

```
$ eCAM_argus_camera
```

When the application is launched, you can view the screen similar to the screen shown below.

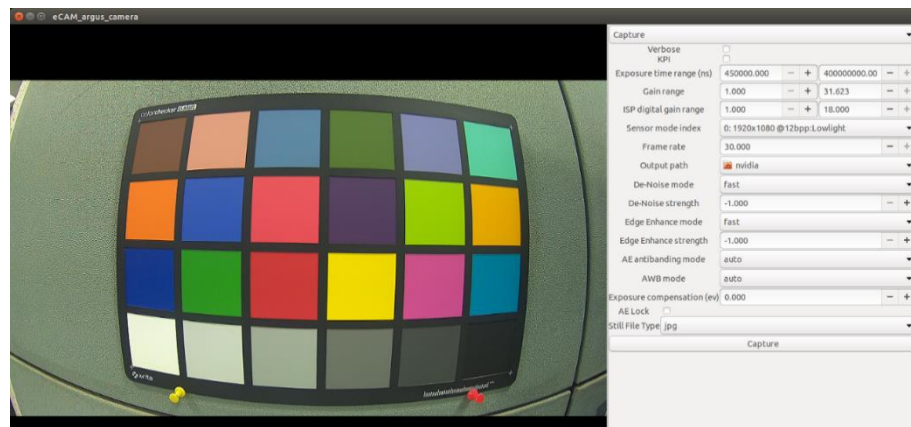


Figure 17: Initial Window Appearance when Application is Launched

Multiple Cameras Streaming:

1. To run the multiple cameras simultaneously, run the following command

```
$ eCAM_argus_camera -d n
```

Note : 'n' denotes the device number. For quad cameras it will be 0-3.

```
nvidia@nvidia-desktop:~$ eCAM_argus_camera -d 0
Executing Argus Sample Application (eCAM_argus_camera)
Argus Version: 0.98.3: (multi-process)
```


Reference Documents

This section describes the software and hardware documents of e-CAM82_CUOAGX. You can download the software and hardware documents from [Developer Resources](#) website.

Software Documents

The software documents and its description are listed in following table.

Table 4: Description of Software Documents

S.NO	What I need	Documents to Refer
1	View the camera stream and change the camera controls using GUI.	e-CAM82_CUOAGX_Linux_App_User_Manual_<REV>.pdf
2	Build and install the GUI based eCAM_argus_camera camera application.	e-CAM82_CUOAGX_eCAM_Argus_Camera_Build_and_Install_Guide_Rev_<REV>.pdf
3	Use gstreamer to control the e-CAM82_CUOAGX camera on the Jetson Orin™/Xavier™ AGX development kit.	e-CAM82_CUOAGX_Gstreamer_Usage_Guide_<REV>.pdf
4	Build custom kernel with support for using e-CAM82_CUOAGX.	e-CAM82_CUOAGX_Developer_Guide_Rev_<REV>.pdf
5	Information about the directory structure and contents of the release package for e-CAM82_CUOAGX.	e-CAM82_CUOAGX-Release_Package_Manifest_<REV>.pdf
6.	Describes the features and specification of e-CAM84_CUMI485C_MOD_H02R1 camera module.	e-CAM84_CUMI485C_MOD_MCU_Protocol_App_Note_Rev_<REV>.pdf

Hardware Documents

The hardware documents and its description are listed in following table.

Table 5: Description of Hardware Documents

S.NO	Documents Name	Description
1	e-CAM82_CUOAGX_Datasheet_<REV>.pdf	Describes the features, connector pin-out details and mechanical dimensions of e-CAM82_CUOAGX.
2	e-CAM84_CUMI485C_MOD_H02R1_Datasheet_<REV>.pdf	Describes the features and specification of e-CAM84_CUMI485C_MOD_H02R1 camera module.
3.	e-CAM82_CUOAGX_Lens_Datasheet_Rev_<REV>.pdf	Describes the optical specification of lenses used in e-CAM82_CUOAGX.

Troubleshooting

In this section, you can view the list of commonly occurring issues and their troubleshooting steps.

I have flashed the Jetson AGX Xavier™ / ORIN™ board with quick start package. After flashing, the board is not booting, or the display is blank. How to solve this issue?

To solve this issue, please try the following:

- Use the correct command with sudo permission whenever required to extract the package.
- Use Ubuntu 18.04 or 20.04 64-bit for flashing.
- Maintain enough free space in hard disk before flashing.

1. How can I get the updated package?

Please login to the [Developer Resources](#) website and download the latest release package.

What's Next?

After understanding the specifications of e-CAM82_CUOAGX board and instructions on how to use this board with NVIDIA® Jetson AGX ORIN™ development kit, you can refer to the following documents to understand more about e-CAM82_CUOAGX.

- *e-CAM82_CUOAGX Developer Guide*
- *e-CAM82_CUOAGX Linux App User Manual*

Glossary

AI: Artificial Intelligence.

API: Application Program Interface.

CMOS: Complementary Metal Oxide Semiconductor.

CSI: Camera Serial Interface.

FPC: Flexible Printed Circuit.

GUI: Graphical User Interface.

MCU: Micro Controller Unit.

MIPI: Mobile Industry Processor Interface.

USB: Universal Serial Bus.

V4L2: Video4Linux2 is a collection of device drivers and API for supporting real-time video capture on Linux systems.

Contact Us

If you need any support on e-CAM82_CUOAGX product, please contact us using the Live Chat option available on our website - <https://www.e-consystems.com/>

Creating a Ticket

If you need to create a ticket for any type of issue, please visit the ticketing page on our website - <https://www.e-consystems.com/create-ticket.asp>

RMA

To know about our Return Material Authorization (RMA) policy, please visit the RMA Policy page on our website - <https://www.e-consystems.com/RMA-Policy.asp>

General Product Warranty Terms

To know about our General Product Warranty Terms, please visit the General Warranty Terms page on our website - <https://www.e-consystems.com/warranty.asp>

Revision History

Rev	Date	Description	Author
1.0	09-Nov-2022	Initial draft	Camera Team
1.1	16-Nov-2022	Added support for Xavier	Camera Team
1.2	22-Feb-2023	4-Lane Support	Camera Team
1.3	10-Mar-2023	Updated software section	Camera Team
1.4	14-Mar-2023	Updated L4T35.2.1 version	Camera Team